Selin Hülagü



PhD Candidate in Operations Analytics

Sustainable Operations | Mathematical Optimization | Supply Chain Analytics

School of Business and Economics, Vrije Universiteit Amsterdam | Amsterdam, the Netherlands

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Academic Profile

I am an operations analyst specializing in quantitative sustainability and mathematical optimization. My research develops decision models and exact algorithms for sustainable supply chains and logistics networks, integrating life-cycle assessment, empirical validation, and real-world operational data. I have been involved in teaching many undergraduate courses, most recently in artificial intelligence, programming, sustainability, and supply chain management, and have supervised both bachelor's and master's thesis projects.

Research Interests

- Quantitative sustainability management, life-cycle assessment and supply-chain optimization
- Exact algorithms, branch-cut-and-price, and mixed-integer mathematical programming
- Sustainable logistics, vehicle routing, and transportation and supply-chain networks
- Machine learning and data-driven decision-making for sustainable operations

Profile Highlights

- Publications in *Transportation Science*, *Journal of Cleaner Production*, *PLOS ONE*, and *IEEE Transactions on Intelligent Transportation Systems*, spanning exact optimization algorithms, sustainable supply chains, and transportation networks.
- University teaching experience since 2017 in operations research, sustainability, supply chains, machine learning, artificial intelligence, and transportation engineering.
- Proven track record in supervising and co-supervising bachelor's and master's thesis research, mentoring students in quantitative analysis, optimization, artificial intelligence, and sustainability-related topics.
- Quantitative research on applied, large-scale sustainability and network problems in international and publicly funded research settings.

Education

PhD Project, Operations Analytics — 2022–Dec. 2026 (expected)
Vrije Universiteit Amsterdam — *Amsterdam, the Netherlands*
Expected defence: March 2027

2025 School on Column Generation — 19–23 May 2025
GERAD; Conservatoire national des arts et métiers (CNAM) — *Paris, France*

Metaheuristics Summer School 2020+1 — 15–18 June 2021
Learning and Optimization from Big Data; University of Catania — *Catania, Italy (virtual)*

MSc, Civil Engineering (Transportation) — 2017–2020
Istanbul Technical University — *Istanbul, Türkiye*
Thesis: Energy Optimized Electric Vehicle Location Routing Problem with Intermediate Nodes

MSc, Civil Engineering (Construction Management) — 2015–2017
Ege University — *Izmir, Türkiye*
Thesis: Experimental and Numerical Investigation of the Effect of Double-Skin Facade Systems on Building Thermal Performance

BSc, Civil Engineering — 2010–2015
Ege University — *Izmir, Türkiye*

Erasmus Exchange, Civil Engineering — 2013
Technische Hochschule Mittelhessen (half-semester exchange) — *Gießen, Germany*

Selected graduate coursework: Graph Theory; Machine Learning; Mathematical Programming; Mathematical and Statistical Methods in Transportation Engineering; Operations Research in Transportation; Intelligent Transportation Systems; Traffic Management; Introduction to Traffic Engineering.

Academic Positions

PhD Candidate, Operations Analytics — 2022–2026
School of Business and Economics, Vrije Universiteit Amsterdam — *Amsterdam, the Netherlands*

Research and Teaching Assistant, Transportation Division — 2017–2022
Department of Civil Engineering, Istanbul Technical University — *Istanbul, Türkiye*

Peer-Reviewed Journal Publications

1. **S. Hülagü**, W. Dullaert, A. S. Eruguz, R. Heijungs, and D. Inghels, "Integrating supply chain life cycle decisions: Empirical model validation for a large-scale case," *Journal of Cleaner Production*, 579, 149547. doi:10.1016/j.jclepro.2026.149547
   Journal AIP: 0.87 — 2026

2. **S. Hülagü**, C. Ciancio, S. Dabia, and W. Dullaert, "An exact algorithm for the vehicle routing problem with time windows and perishable products," *Transportation Science*, Articles in Advance. doi:10.1287/trsc.2025.0262
   Journal AIP: 0.91 — 2026

3. **S. Hülagü**, W. Dullaert, A. S. Eruguz, R. Heijungs, and D. Inghels, "Integrating life cycle assessment into supply chain optimization," *PLOS ONE*, 20(1), e0316710. doi:10.1371/journal.pone.0316710
   Journal AIP: 0.67
   Citation-based AIP: 0.92 — 2025

4. **S. Hülagü** and H. B. Çelikoğlu, "Electric vehicle location routing problem with vehicle motion dynamics-based energy consumption and recovery," *IEEE Transactions on Intelligent Transportation Systems*, 23(8), 10275–10286. doi:10.1109/TITS.2021.3089675
   Journal AIP: 0.89
   Citation-based AIP: 0.92 — 2022

5. **S. Hülagü** and H. B. Çelikoğlu, "Environment-friendly school bus routing problem with heterogeneous fleet: A large-scale real case," *IEEE Transactions on Intelligent Transportation Systems*, 23(4), 3461–3471. doi:10.1109/TITS.2020.3036696
   Journal AIP: 0.89
   Citation-based AIP: 0.66 — 2022

6. **S. Hülagü** and H. B. Çelikoğlu, "An electric vehicle routing problem with intermediate nodes for shuttle fleets," *IEEE Transactions on Intelligent Transportation Systems*, 23(2), 1223–1235. doi:10.1109/TITS.2020.3023673
   Journal AIP: 0.89
   Citation-based AIP: 0.90 — 2022

Manuscripts Under Review

1. **S. Hülagü**, W. Dullaert, S. Dabia, and M. Hoogeboom, "Branch-cut-and-price for the vehicle routing problem with multiple time windows." Under review at *Transportation Science*.
   Journal AIP: 0.91

Research Experience and Funded Projects

Researcher, Intelligent Transportation Systems Research Lab — 2017–2022
Istanbul Technical University — *Istanbul, Türkiye*

Project Researcher — 2019–2022
Environment Friendly Optimization of Transportation Networks: Analysis Over Small Sized Real Network — *Istanbul, Türkiye*
Istanbul Technical University; Scientific and Technological Research Council of Türkiye, Project No. 218M307.

Project Researcher — 2016–2017
HIT2GAP: Highly Innovative building control Tools Tackling the energy performance GAP — *Izmir, Türkiye*
Ege University; European Union Horizon 2020; Grant Agreement No. 680708.

Erasmus Research Intern (Undergraduate Researcher) — 2014, three months
Fraunhofer Institute for Building Physics (IBP) — *Kassel, Germany*

Conference and Book Contributions

1. M. A. Silgu, I. G. Erdağı, **S. Hülagü**, S. Akti, H. Akova, G. Akyol, S. Göncü, G. Göksu, and H. B. Çelikoğlu, "A methodology to consider explicitly emissions in dynamic user equilibrium assignment," in *Computer Aided Systems Theory–EUROCAST 2022*, Lecture Notes in Computer Science, vol. 13789, Springer, pp. 362–369. doi:10.1007/978-3-031-25312-6_42 — 2022

2. H. Akova, **S. Hülagü**, and H. B. Çelikoğlu, "Effects of energy consumption on cost optimal recharging station locations for e-scooters," in *2021 7th International Conference on Models and Technologies for Intelligent Transportation Systems (MT-ITS)*, IEEE. doi:10.1109/MT-ITS49943.2021.9529282 — 2021

3. **S. Hülagü** and H. B. Çelikoğlu, "Electrified location routing problem with energy consumption for resources restricted archipelagos: Case of Buyukada," in *2020 Forum on Integrated and Sustainable Transportation Systems (FISTS)*, IEEE, pp. 323–327. doi:10.1109/FISTS46898.2020.9264865 — 2020

4. **S. Hülagü** and H. B. Çelikoğlu, "A mixed integer linear programming formulation for green vehicle routing problem: Case for shuttle services," in *Computer Aided Systems Theory–EUROCAST 2019*, Lecture Notes in Computer Science, vol. 12014, Springer, pp. 153–160. doi:10.1007/978-3-030-45096-0_19 2020

5. **S. Hülagü**, G. Atasavar, and H. B. Çelikoğlu, "Green routing plan for university shuttle services using mixed integer linear programming," in *2019 IEEE 5th International Forum on Research and Technologies for Society and Industry: Innovation to Shape the Future (RTSI)*, pp. 471–476. doi:10.1109/RTSI.2019.8895571 2019

6. **S. Hülagü** and H. B. Çelikoğlu, "A multiple objective formulation of an electric vehicle routing problem for shuttle bus fleet at a university campus," in *2019 6th International Conference on Models and Technologies for Intelligent Transportation Systems (MT-ITS)*, IEEE. doi:10.1109/MTITS.2019.8883344 2019

7. **S. Hülagü** and H. B. Çelikoğlu, "An integer linear programming formulation for routing problem of university bus service," in *New Trends in Emerging Complex Real Life Problems*, AIRO Springer Series, vol. 1, Springer, pp. 303–311. doi:10.1007/978-3-030-00473-6_33 2018

Teaching Experience (Tutorials, exam and course support)

Vrije Universiteit Amsterdam *Amsterdam, the Netherlands*

Programme: BSc Creative Technology (joint programme with University of Twente)
- Programming for Artificial Intelligence 2024–present
- Artificial Intelligence Theory 2024–present

Programme: Minor programme Sustainability: Management and Innovation
- Sustainable Business Processes 2022–present

Programme: BSc International Business Administration
- Global Sustainability 2022–present

Programme: Minor programme E-Business and Online Commerce
- E-Commerce Supply Chain Management 2022–2024

Istanbul Technical University *Istanbul, Türkiye*

Programme: MSc Transportation Engineering
- Intelligent Transportation Systems 2019–2022
- Operations Research in Transportation 2022

Programme: BSc Civil Engineering
- Highway Engineering 2017–2022
- Railway Engineering 2017–2022

Student Supervision

MSc theses

1. **Philippe van Mastrigt**, "Supply Chain Life Cycle Optimization under Uncertainty." 2026
   MSc Embedded Systems, TU Delft — Co-supervisor

2. **Yu An Chen**, "The Effect of Resource Sufficiency Strategies in Circular Supply Chains: A System Dynamics Approach." 2024
   MSc Transport and Supply Chain Management, Vrije Universiteit Amsterdam — Co-supervisor

3. **Afsaneh Rahmani**, "Optimization of Risk and Cost in the Inventory Routing Problem with Time Windows for Hazardous Materials." 2023
   MSc Transport and Supply Chain Management, Vrije Universiteit Amsterdam — Co-supervisor

4. **Thi Phuong Ly Nguyen**, "The environmental impacts of grocery retail before vs after integrating online business: The choice of online fulfilment strategies." 2022
   MSc Transport and Supply Chain Management, Vrije Universiteit Amsterdam — Co-supervisor

5. **Daan Rutten**, "Modern Slavery in the Lower Tiers of the Supply Chain: An Analysis of the Effect of the "Decent Work" Principle from the Better Cotton Initiative." 2022
   MSc Transport and Supply Chain Management, Vrije Universiteit Amsterdam — Co-supervisor

BSc theses

1. **Luca Barbulescu**, "Decision-Making in Global Perishable Humanitarian Logistics: A Systematic Literature Review of Speed, Product Quality, and Cost Trade-Offs." 2026
   School of Business and Economics, Vrije Universiteit Amsterdam — Main supervisor

2. **Michele Chang**, "Distribution Structure and Route Availability in Humanitarian Logistics: An International Review of Delivery Performance for Perishable Goods under Disrupted Infrastructure." 2026
   School of Business and Economics, Vrije Universiteit Amsterdam — Main supervisor

3. **Wouter Tol**, "Analyzing the Effect of Congestion Pricing Policies: Evaluation of Both Theoretical and Case Studies." 2026

School of Business and Economics, Vrije Universiteit Amsterdam — Main supervisor

4. **Dania Toumi**, "Electric Vehicle Integration in Urban Last-Mile Delivery: A Comparative Literature Review of East Asia and Western Europe." 2025
   School of Business and Economics, Vrije Universiteit Amsterdam — Main supervisor
5. **Lucas Twigt**, "A Systematic Literature Review on the Challenges and Benefits of Combining Life Cycle Assessment and Sustainable Supply Chain Management." 2024
   School of Business and Economics, Vrije Universiteit Amsterdam — Main supervisor
6. **Kelvin Kramer**, "Sustainable Supply Chain: Assessing the Three Pillars of Sustainability and Enhancing Supply Chain." 2024
   School of Business and Economics, Vrije Universiteit Amsterdam — Main supervisor

Professional Service and Reviewer Activities

Reviewer: *IEEE Transactions on Intelligent Transportation Systems*; *IEEE Intelligent Transportation Systems Magazine*.

Methods and Technical Skills

Research methods: mixed-integer linear programming; exact algorithms; branch-cut-and-price; column generation; multi-objective optimization; life-cycle assessment; machine learning; empirical model validation.
Teaching areas: artificial intelligence; machine learning; programming; sustainability; supply-chain management.

Languages

Turkish	Native	**English**	Full professional proficiency
Dutch	A2 level (VU Amsterdam language course)	**French**	B1 level (Institut Français Istanbul course)